

swtpcsim — FLEX9 on the SWTPC 6809

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The built-in `swtpc09` machine is an SWTPC 6809: the MP-09 processor board with the S-BUG monitor, 56K of RAM, an MP-S console, a DC-4 floppy controller (`dc4`) and the MP-ID interface driver (`mpid`), whose 6840 is FLEX9's timer. Each machine file here is that machine with a TSC **FLEX9 2.8:3** boot disk mounted on drive 0, so it comes up at S-BUG's `>` prompt ready to boot:

```
swtpcsim examples/flex9/flex9-40.toml
S-BUG 1.8 - 56K
>U                               (S-BUG mini-floppy boot)
```

`U` is S-BUG's mini-floppy boot command. It restores drive 0, reads track 0 sector 1 into `$C000` and jumps there. FLEX signs on, asks the date, and drops to its `+++` prompt:

```
FLEX - Version 2.8:3 - 56K

Date (MM,DD,YY)? 1,15,80

+++CAT
```

From `+++`, `CAT` lists the disk and the usual FLEX commands work. `TIME` runs a command and says how long it took, counted on the MP-ID's timer:

```
+++TIME CAT
...
-- Elapsed time was      2.78
```

The time is emulated seconds: flat out, the machine runs faster than a real 6809. Set the CPU's `clock_hz` in the `.toml` for seconds that match the wall. A machine without the MP-ID still runs FLEX, which then says `Timer not available.` at boot.

The three disks

The same FLEX9 in the three geometries the DC-4 supports. The board finds the geometry from the image's **size**, so nothing in a `.toml` sets tracks or sides:

Machine file	Disk	Geometry
<code>flex9-35.toml</code>	<code>FLEX9-2.83-35.DSK</code>	35-track single-sided, 88K
<code>flex9-40.toml</code>	<code>FLEX9-2.83-40.DSK</code>	40-track single-sided, 100K
<code>flex9-DS.toml</code>	<code>FLEX9-2.83-DS.DSK</code>	40-track double-sided, 200K

All three come from Mike Douglas's *PC2Flop and Flop2PC (for SWTPC DC-x Controllers in SWTPC 6809)* on deramp.com:

<https://deramp.com/downloads/swtpc/software/FLEX/6809%20FLEX/Disk%20Image%20Transfer/>

Read-only by default

Drive 0 is mounted `wriTEprotect = true`, so looking around FLEX can never change the image. To save files back (FLEX's `SAVE`, `NEWDISK`, a file copy), remove the `wriTEprotect` line from the drive in the `.toml`, or copy the `.DSK` somewhere writable first.